

Deriving I2SB as Annealed Conditional-Tweedie Ascent

A corrected derivation for pure image-to-image translation (T1 -> T1CE)

We translate a source y (domain A, e.g. T1) into a target x (domain B, e.g. T1CE), from paired data $(x, y) \sim p(x, y)$ and with no measurement operator. This note derives the I2SB algorithm as an annealed ascent driven by a conditional-Tweedie residual, and isolates the one substitution that separates a blurry one-shot regressor from the bridge.

1. The MAP objective, and why the Bayes split fails

The translation MAP problem is

$$x^\star = \arg \max_x p(x|y)$$

The usual move is Bayes, $\arg \max_x [\log p(y|x) + \log p(x)]$, which needs an analytic likelihood $p(y|x)$. That exists only when y is a measurement of x through a known operator. In pure translation the whole inter-domain relationship is what we are learning, so a Gaussian stand-in

$$p(y|x) = \mathcal{N}(y; x, \tau) \Rightarrow \log p(y|x) = -\frac{1}{2\tau} \|x - y\|_2^2$$

is not an approximation but a different problem (restoration): it pulls x toward y , biasing the synthesized T1CE back toward T1. We therefore drop the factorization and build a path of densities between the two domains instead.

2. A homotopy between the two data densities

Introduce a stochastic interpolant between target and source, indexed by $t \in [0, 1]$:

$$x_t = \mu_0(t)x + \mu_1(t)y + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I),$$

with $\mu_0(0) = 1$, $\mu_1(0) = 0$ (so $x_0 = x$) and $\mu_0(1) = 0$, $\mu_1(1) = 1$ (so $x_1 = y$), and σ vanishing at both ends. Its marginals p_t form a continuous homotopy from $p_0 = p(x)$ to $p_1 = p(y)$. The Brownian-bridge choice of (μ_0, μ_1, σ) recovers I2SB exactly; any other valid interpolant yields a sibling method (flow / bridge matching).

3. Conditional Tweedie on the bridge variable (the correction)

The reverse dynamics carrying $p(y)$ back to $p(x|y)$ have a single unknown, the denoiser

$$D(x_t, t) = \mathbb{E}[x|x_t],$$

estimated by plain regression $\min_{\theta} \mathbb{E} \|D_{\theta}(x_t, t) - x\|_2^2$. This is the conditional-Tweedie object, done correctly. Given the pairing,

$$x_t | (x, y) \sim \mathcal{N}(\mu_0 x + \mu_1 y, \sigma^2 I),$$

so x_t is a genuine Gaussian perturbation of x , and the residual is a valid score of the time- t marginal:

$$D(x_t, t) - x_t \propto \nabla \log p_t(x_t)$$

Contrast the tempting but wrong residual $\mathbb{E}[x | y] - y$: since y is not a Gaussian corruption of x , it is the score of nothing, and it is static -- one fixed vector, with nothing to anneal. The valid residual is evaluated at the evolving iterate x_t , not at y .

4. The annealed update is the ImMAP relaxation step

Discretizing the reverse dynamics gives the update

$$x_{t-1} = x_t + h_t(D(x_t, t) - x_t) + \gamma_t \varepsilon, \quad x_{t_{\max}} = y,$$

a relaxation toward the denoiser output plus annealed noise. It is structurally identical to the ImMAP solver

$$x_t = x_t + h_t(v_t - x_t) + \gamma_t \varepsilon,$$

with the two differences that make translation differ from reconstruction: (i) the denoiser $v_t = D(x_t, t)$ is trained on the bridge marginals rather than on Gaussian noise, and (ii) the source y is absorbed into D through the interpolant, so no measurement operator or data-consistency solve ever appears.

5. $\mathbb{E}[x | y]$ is exactly the first step

Your instinct was almost right: the very first iterate uses

$$x_{t_{\max}} = y, \quad D(y, t_{\max}) \approx \mathbb{E}[x | y],$$

so $\mathbb{E}[x | y] - y$ is precisely the first step of I2SB -- the blurry MMSE mean. The algorithm does not stop there: it re-evaluates D on the refined iterate x_{t-1} and repeats. That re-evaluation is the anneal, self-consistent only because the bridge trained D on the whole path, not just at $x_t = y$. In one line: I2SB applies the paired regressor to its own progressively refined output, instead of once to y .

6. MMSE vs. MAP (a bookkeeping caveat)

Tweedie yields the posterior mean $\mathbb{E}[x|x_t]$, not the mode, so the stochastic iteration samples $p(x|y)$ rather than solving $\arg \max$. The two agree only for symmetric posteriors. For a single deterministic estimate -- the honest stand-in for MAP -- take the OT-ODE limit $\gamma_t \equiv 0$, a deterministic transport $y \mapsto x$. Keep the noise for diverse samples; drop it for one canonical TICE (often the safer clinical choice, given the risk of hallucinated enhancement).

Algorithm (stochastic I2SB sampling)

```
initialize  x = y
for t = t_max, ..., 1:
    x_hat = D(x, t)                # paired regressor at the CURRENT iterate
    x      = x + h_t (x_hat - x) + gamma_t * noise
return x      # a sample of p(x|y); set gamma_t = 0 for the MAP-like transport
```